Expanding Quantum Hall Simulation

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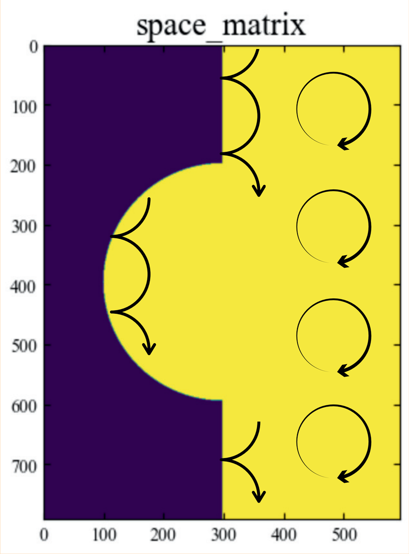
Importance

Expanding Quantum Hall edges act very similar to an expanding universe. If we can control that expansion, we can simulate redshift from the early expanding universe, general relativity, black holes and more.

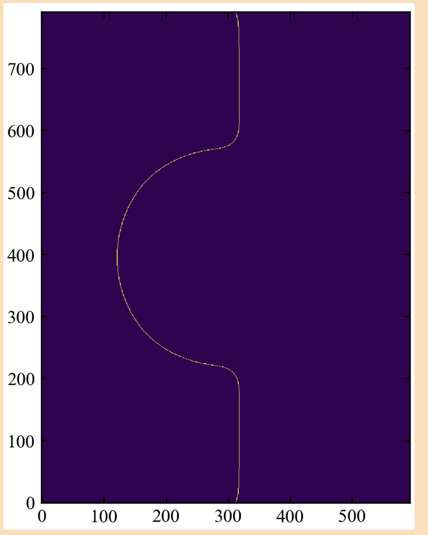
Background and Setup

Quantum Hall effect



- GaAs wafer.
- Etched to the yellow shape.
- Apply a magnetic field.
- Electrons can only move in circles.
- Electrons along the edge bounce and can travel.

Edge definition



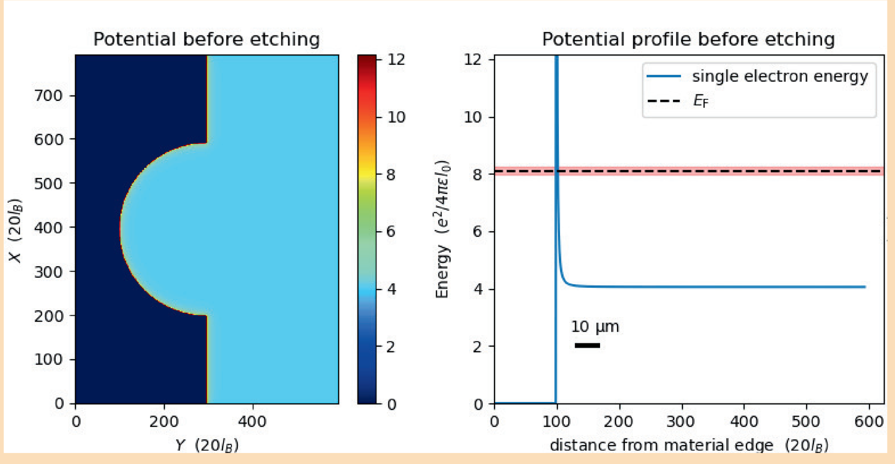
- We approximate the edge to be where $|U - E_{Fermi}| < E_{fluc}$
- This is the region where electrons can move.
- For a precise edge position we use where $U = E_{fermi}$

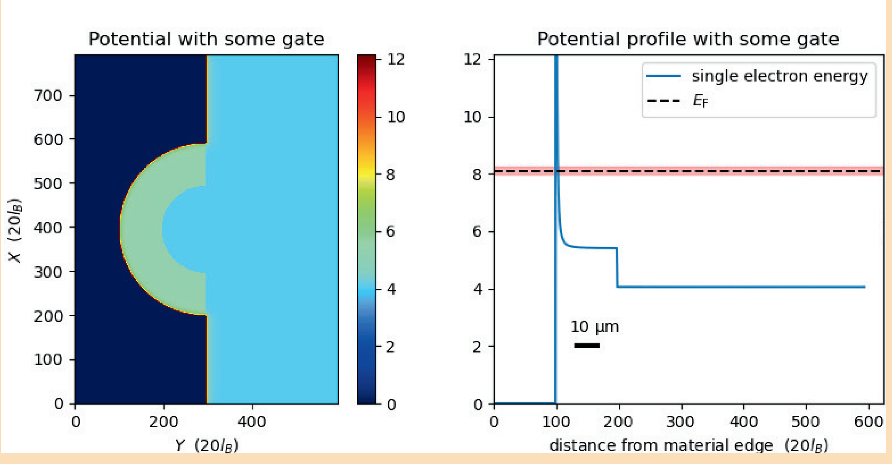
Energy Structure

- Electrons can't leave material.
- We model as potential $U(x) \sim \frac{1}{x^2}$
- We apply a minimum ground state energy.

Gate

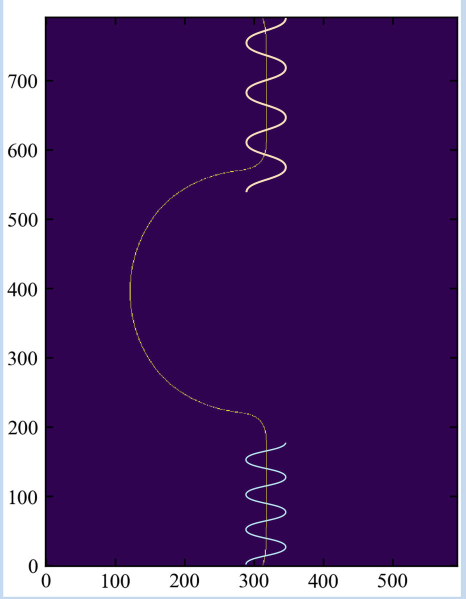
- We apply a voltage to the circular region.
- The potential increases.
- The intersection of the energy with the Fermi energy moves farther out.
- The edge shrinks.





Measuring Redshift

We want to make a preliminary model that measures redshift and thus confirm edge expansion occurred.



- Create an electron pulse along Quantum Hall edge.
- Expand/shrink the edge as the pulse is traveling along it.
- Make sure the pulse is fully in the expanding region when the expansion/shrinking begins.
- Make sure the front of the pulse stays in the expanding region the whole time.
- Measure the frequency of the shifted pulse.

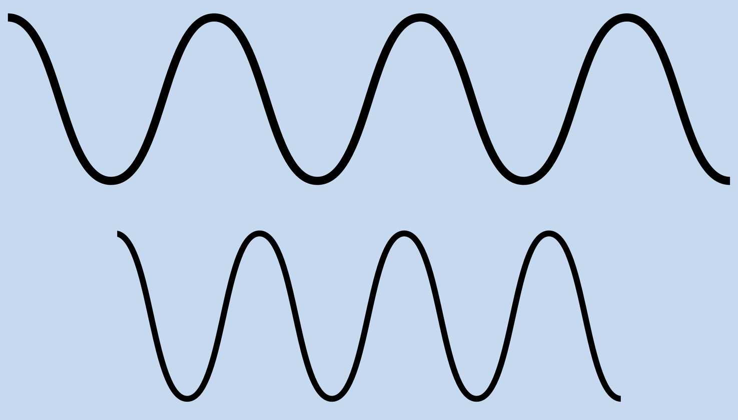
Scale Factor Blueshift

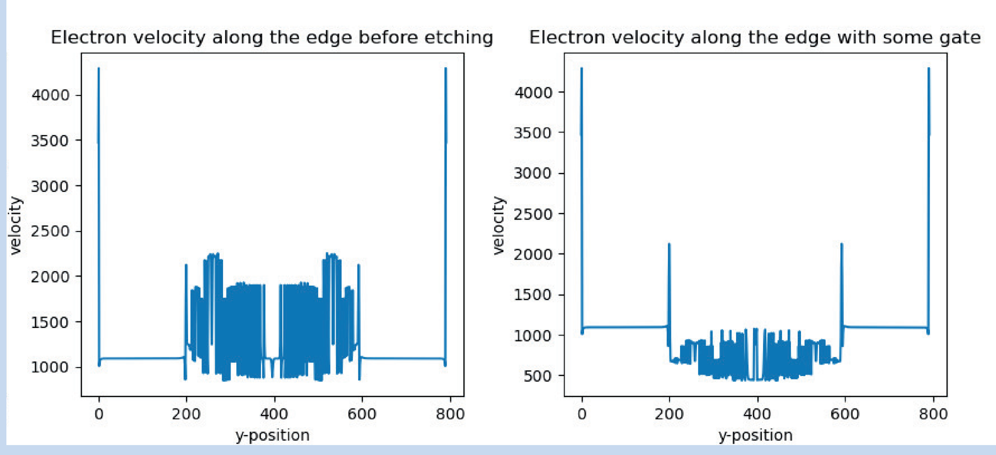
- Edge length decreases.
- The distance between nodes of an electromagnetic wave decreases.
- Velocity remains constant.
- Therefore frequency increases.

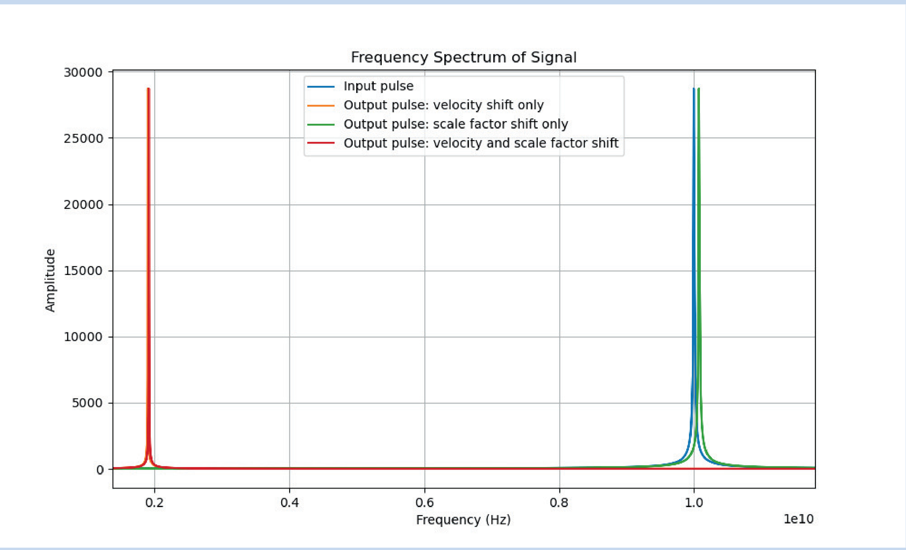
$$f = \frac{v}{\lambda}$$

Velocity Redshift

- Velocity decreases.
- The distance between nodes of an electromagnetic wave remains constant.
- Therefore frequency decreases.





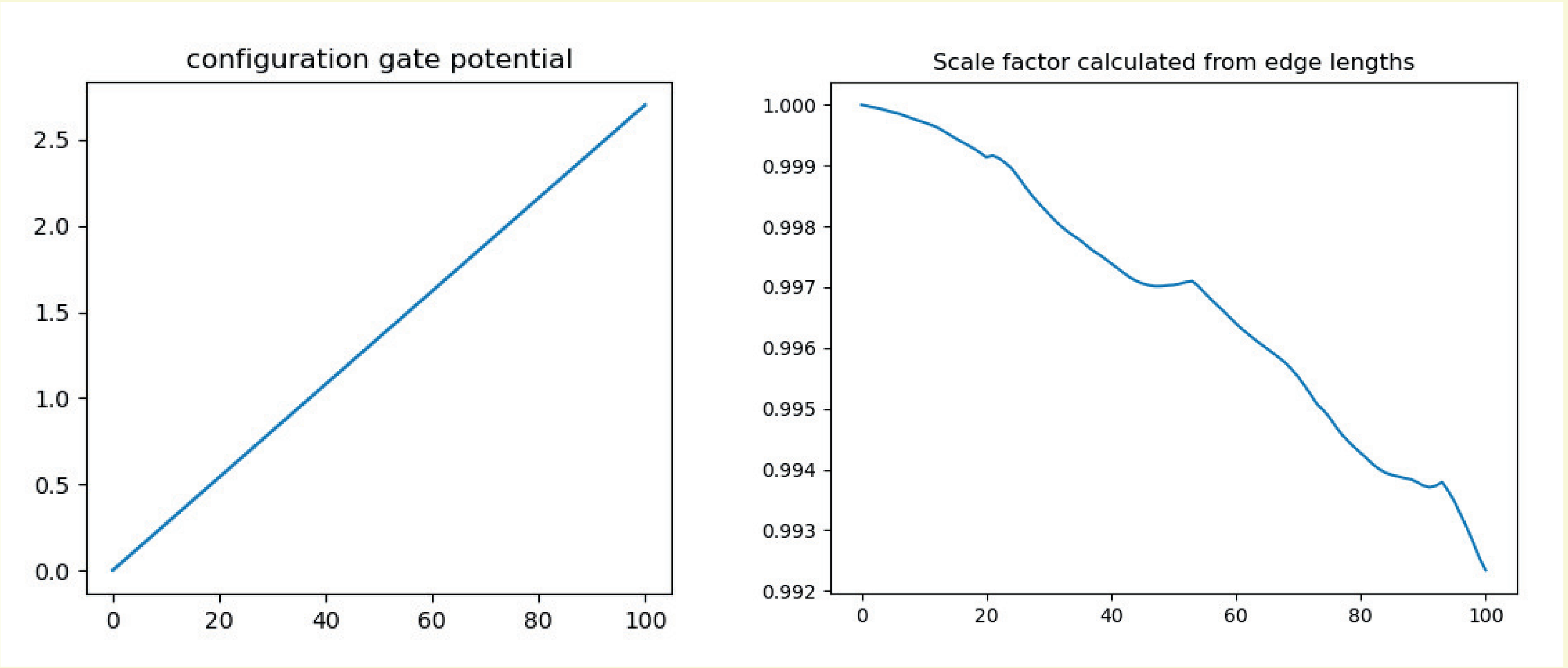


Results:

- Velocity changes the frequency much more than scale factor.
- Both changes are detectable.

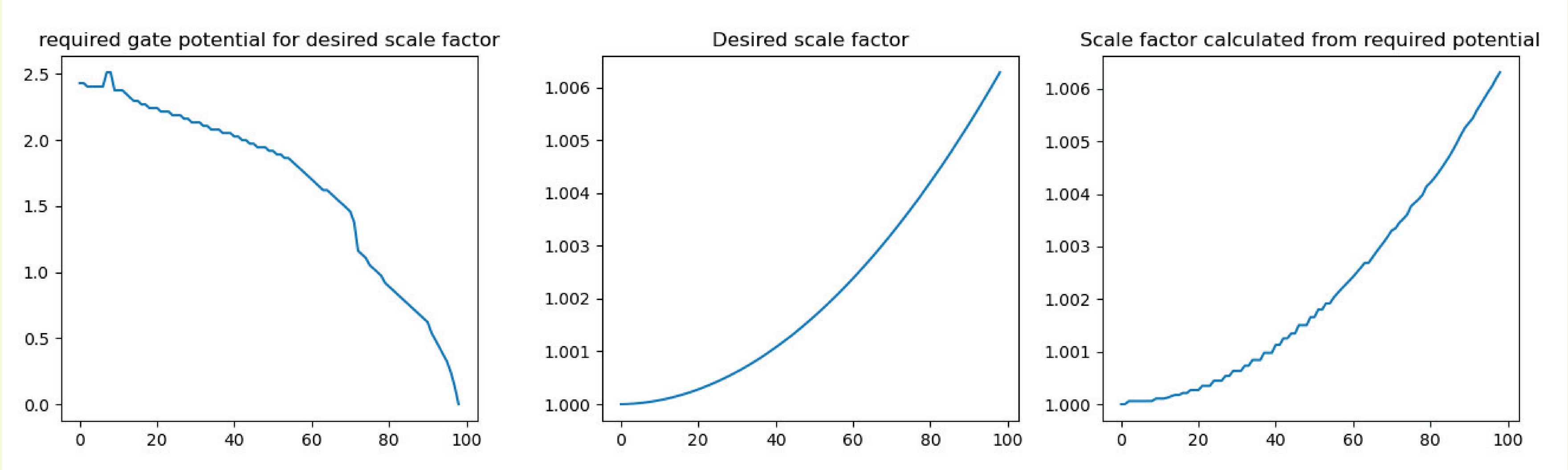
Calculating Scale Factor

- Scale factor is the ratio of edge length to initial edge length.
  - Only defined for uniform expansion
  - Theory says the expansion should be uniform
- Find the edge, calculate length for each gate voltage



Generating a scale factor

- We know edge length at each gate voltage.
- Apply the gate voltage that gets closest.
- Cosh^2 (t) scale factor models expanding universe
  - We want to be able to generate that

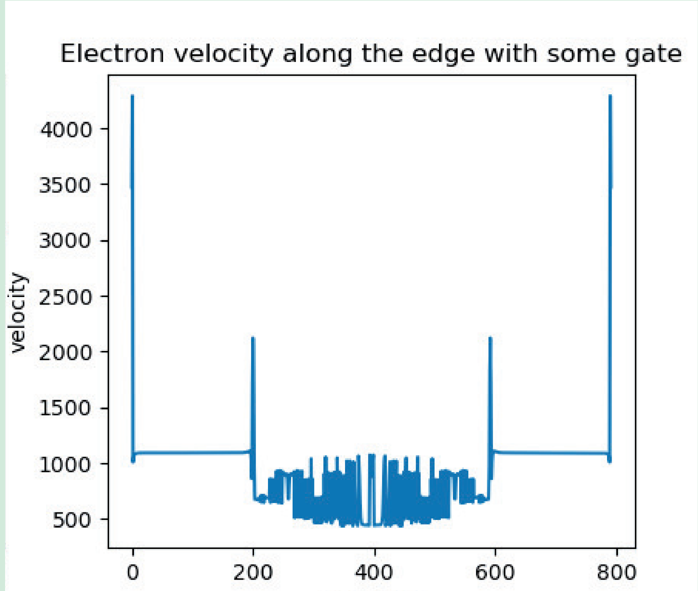


Dynamics

Velocity

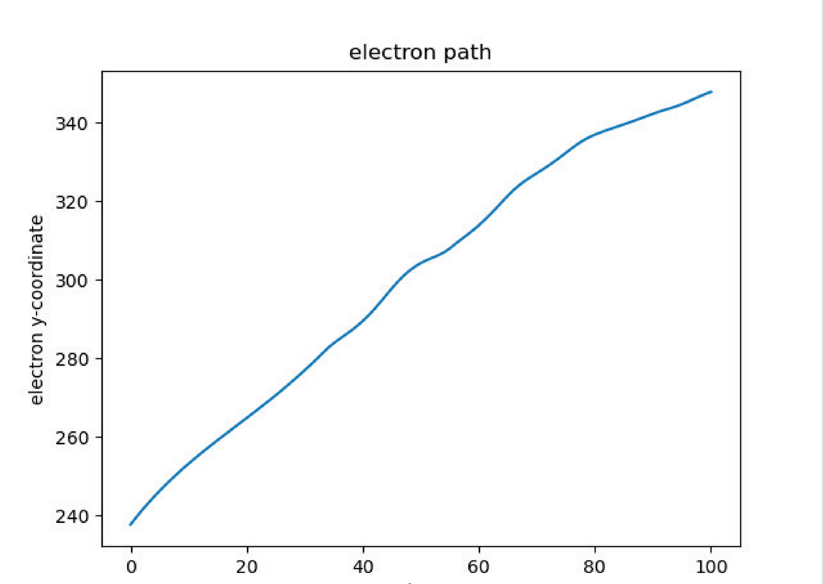
$$v = \frac{|\nabla E|}{B \cdot e}$$

- Theory says it should be uniform.
- Noise comes from pixelation.



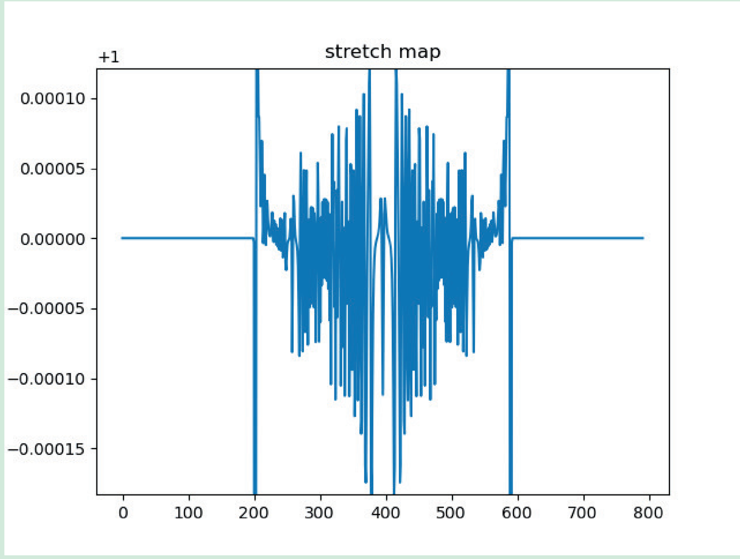
Path

- We know the velocity.
- We know where the edge is.
- Numerically solve the differential equation to find the position.



Dynamic Curvature

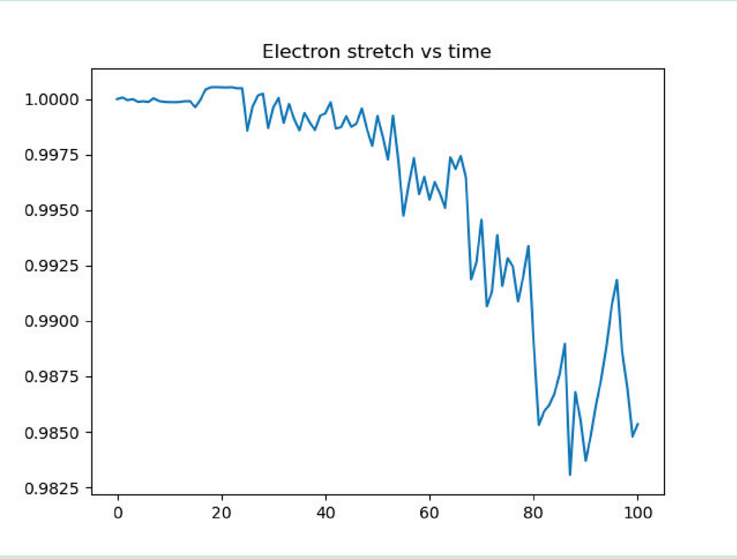
- Compute the ratio of the marginal edge length to the marginal edge length at the next time step at each point.
- Very noisy - we didn't use it for much else.
  - Noise comes from pixelation.
  - Theory says it should be uniform.



Scale Factor 2.0

- Producting the curvature tells us the ratio of the new marginal edge length to the initial.
- Producting along a path gives the scale factor experienced by an electron.
- Compute via logarithm transform.

$$\ln\left(\prod_{k=1}^{N} s(t(k))^{t(\Delta k)}\right) = \sum_{k=0}^{N} t(\Delta k)\ln(s(t(k))) \to \int_0^{t_0} \ln(s(t))dt$$



Future prospects include configurations to see a larger change in scale factor. The simulation has strong adaptability, making it ideal to simulate the effects of many different experimental setups, including a position-dependent gate voltage and shallow etchings to create slope in the energy profile to keep edge width down as it moves farther from the material edge. Additional experiments could measure the Hubble constant of this expanding universe as well as simulate the event horizon of a black hole to measure Hawking radiation.

With assistance from Professor Go Yusa, Assistant Professor Nicholas Moore, and Grad Student Yunhyon Jeong